

Lesson 8: Accessing earth observations (EO) datasets

Accessing EO data via web-based portal

EARTHDATA SEARCH Find a DAAC

Feedback 3 ?

Q modis surface reflectance | **Search** Search Results (77 Collections)

Temporal Spatial  

Spatial Rectangle 

SW: 2.51902,-23.06249
NE: 20.08846,-4.78124

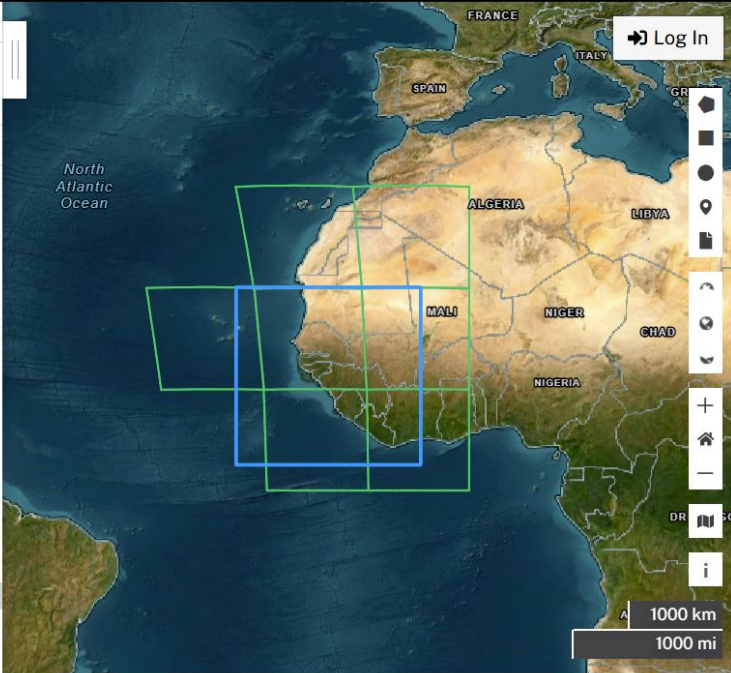
Temporal
Start
2025-01-01 00:00:00
YYYY-MM-DD HH:mm:ss
End
2025-01-07 23:59:59
YYYY-MM-DD HH:mm:ss
☐ Use a recurring date range

Day/Night

Showing 14 of 14 matching granules Sort: Start Date (Newest) View: List

MOD09A1.A2025001.h16v06.061.2 025011010944	 START 2025-01-01 00:00:00 END 2025-01-08 23:59:59 + Download	MOD09A1.A2025001.h17v07.061.2 025011010934	 START 2025-01-01 00:00:00 END 2025-01-08 23:59:59 + Download
MOD09A1.A2025001.h16v07.061.2 025011010824	 START 2025-01-01 00:00:00 END 2025-01-08 23:59:59 + Download	MOD09A1.A2025001.h16v08.061.2 025011010922	 START 2025-01-01 00:00:00 END 2025-01-08 23:59:59 + Download
MOD09A1.A2025001.h17v08.061.2		MOD09A1.A2025001.h17v06	

[Add](#) [Download All](#) 14 Search Time: 0.2s


Log In

1000 km
1000 mi

MODIS/Terra Surface Reflectance 8-Day L...

0 2000 2010 2020 2030

v26.1.4-5 · NASA Official: Doug Newman · FOIA · NASA Privacy Policy · USA.gov

Earthdata Access: A Section 508 accessible alternative

Accessing EO data via API



earthaccess

A Python Library for NASA Earthdata

```
1 import earthaccess
2
3 earthaccess.login(strategy="netrc")
4
5 results = earthaccess.search_data(
6     count=2,
7     short_name="ATL08",
8     bounding_box=(-92.86, 16.26, -91.58, 16.97),
9 )
10
11 earthaccess.download(results, "./research/")
12
```

- Line 3: earthaccess handles authentication with NASA EDL.
- Line 5: earthaccess abstracts NASA's search API (CMR) into a *pythonic* module.
- Line 11: earthaccess can download or open data for both cloud and on-prem hosted datasets.

Other EO data access options

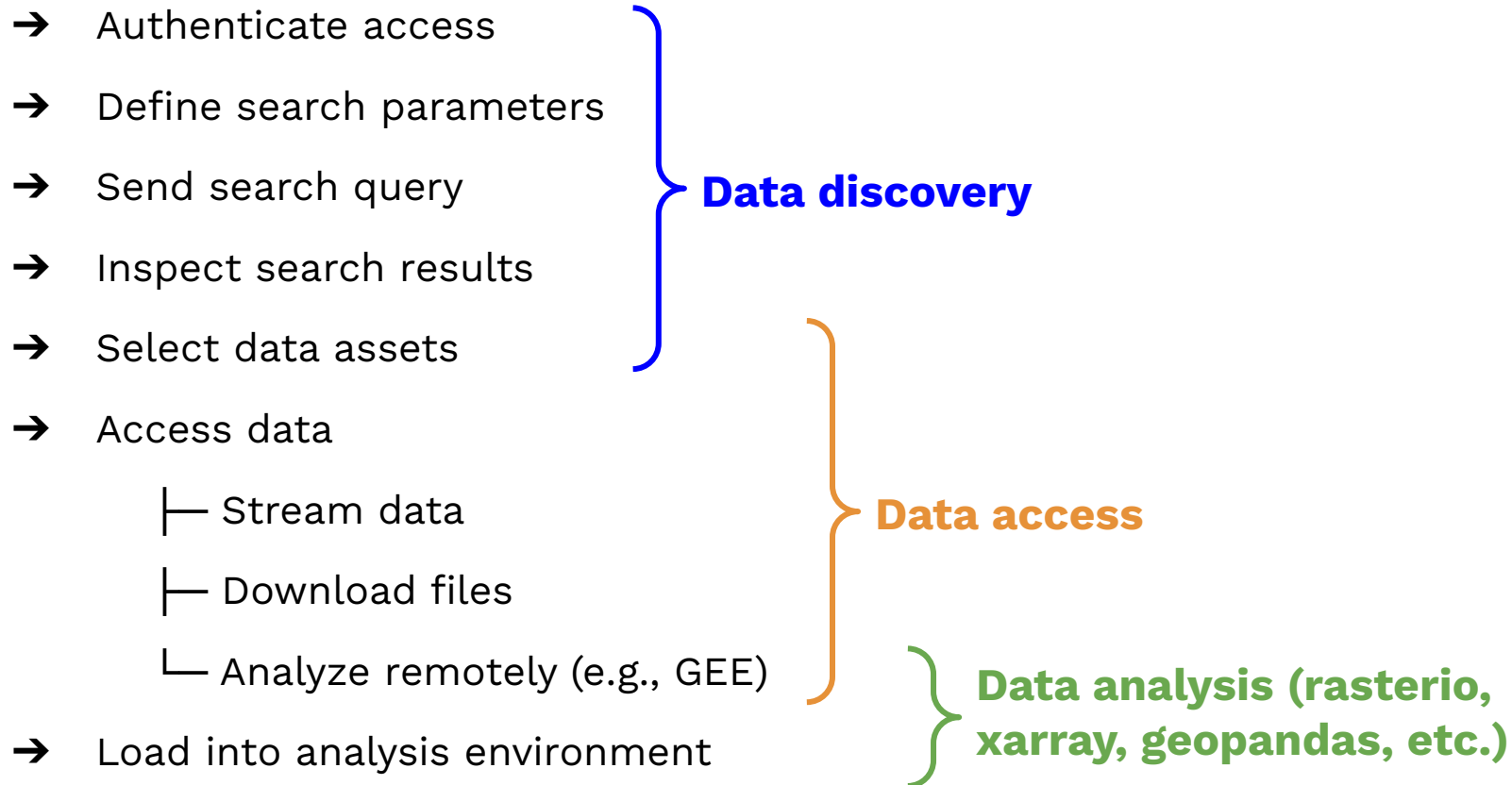
Name	Description	Python libraries	Examples
NASA earthaccess	Access NASA EarthData products	earthaccess	MODIS, Landsat, ICESat-2
Earth Observation Data Centre (EODC)	Access EU Copernicus / ESA products	pystac-client	Sentinel-2, Sentinel-1 Global Flood Monitoring, C3S soil moisture
Google Earth Engine (GEE)	These access options often overlap in the datasets they contain, but differ in where the data is stored and the online processing options		
Microsoft Planetary Computer	analyze cloud datasets		Dynamic World LULC, AIP, USDA Cropland Data Layer
Amazon Web Server (AWS) S3	Cloud storage API for specific datasets	boto3	MayMet, ERA5, Microsoft Building Footprints
			NOAA-GMU VIIRS Flood Detection, CYGNSS



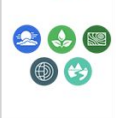
Microsoft Planetary
Computer



Typical EO data access API workflow



This guide highlights some of the data tools available for discovering, browsing, and accessing NASA Earth science data. By knowing common tasks the tools can accomplish, you'll have more time to process data and maximize scientific return.

KEY		TOPICS										
		 User can achieve this task within this tool.	 Task can be partially completed with this tool.	 Task cannot be completed within this tool alone.	 Atmosphere	 Biosphere	 Climate Indicators	 Cryosphere	 Human Dimensions	 Land Surface	 Ocean	 Solid Earth
DATA TASKS		Worldview	Giovanni	Earthdata Search	CMR Search API	FIRMS	AppEEARS	VEDA Dashboard	OPeNDAP	EGIS	Panoply	
	Explore/Find Data											
	Co-locate variables ¹											
	Narrow Down/Subset Data → Subset Spatially for a Point											
	Narrow Down/Subset Data → Subset Spatially for an Area											
	Access/Get Data → Download data directly											
	Customize/Transform Data → Reformat data											
	Analyze Data → Analyze in-browser											
	Analyze Data → Correlate variables											
	Analyze Data → Create time series											
	Analyze Data → Detect Change											
	Analyze/Visualize Data → Visualize primary variables											
	Application Programming Interface (API) available											
	Download requires Earthdata Login											
	GIS-Friendly Output/Integration											
Programming skills required												
 Includes entire NASA Earth science data archive												
 Web-based  Local install												
 Browse  Download ²		 	 			 			 			
1. A variable is any of a set of physical properties whose values determine the characteristics or behavior of something. For example, variables of the atmosphere include temperature, pressure, and density. 2. Download includes downloading subsets of data.	Earthdata Topic Areas 										N/A	

1. A variable is any of a set of physical properties whose values determine the characteristics or behavior of something. For example, variables of the atmosphere include temperature, pressure, and density.

2. Download includes downloading subsets of data.

Considerations when choosing data access tools

- How much **time** do I have / when do I need my output by?
- What is my **intended output and quality**? → E.g., Map, single image, time-series, long-terms statistics, folder of files, data cube in netcdf, etc.
- Do I need to **scale** up? → I.e., spatially, temporally, by collection?
- Do I need **instant visualization** options?
- Do I need to be able to **manipulate data values**? E.g., scaling surface reflectance values.
- Do I need to use the data **values as inputs to a calculation**? E.g., NDVI.
- Do I need to perform **processing on the fly**? E.g., merging, mosaicing, reprojecting.

EO data access activity

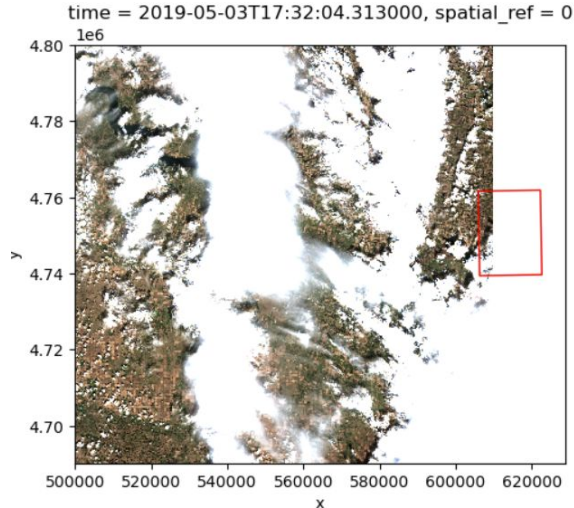
1. **Choose one “download data” code** from the snippets folder in GitHub—pick the method that is most relevant to your work or interests you most.
2. **Find a dataset** you want to download and define your own region and start and dates.
3. **Perform a data search** and inspect the results. Bonus: play around with the search parameters—e.g., is it possible to filter by cloud cover? (Read the docs).
4. Stream the data and **create a plot**.
5. **Download the data** as a tif or netcdf file.

<https://github.com/NASA-SARP/2026-Coding-Alex/blob/main/snippets/>

snippets

This folder contains general code snippets.

- [Download data from an AWS bucket](#)
- [Download data from NASA EarthData using earthaccess](#)
- [Download data from Copernicus EODC using STAC API](#)
- [Download data from Google Earth Engine](#)



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Please complete the **exit ticket**:

<https://forms.gle/UuvpwSEoqDbV5oGo9>

Don't forget to **shut down CryoCloud**:

File → Hub Control Panel → Stop Server